FRONTEND ENGINEERING ASSESSMENT

Joineazy

Student, Group & Assignment Management System

Round 2 Complete Production Implementation

Live Deployment:	https://frontend-v3yg.vercel.app
GitHub Repository:	https://github.com/214G1A05C2/joineasytask2.git
Role Evaluation:	Professor & Student Role-Based Access
Architecture:	React 18 + Vite 6 + Tailwind CSS + Context API + Recharts

Status: 100% Production Ready & Tested

Date: November 2026

1. Executive Summary

Comprehensive Project Overview & Assessment Deliverable

Joineazy is a production-grade, highly responsive university productivity platform engineered to streamline assignment workflows, collaborative team formations, and submission tracking between professors and students.

> **Key Milestone Achievement:** Built from ground up following strict SaaS architectural patterns. The platform includes full client-side routing, JWT session persistence via localStorage, reactive data synchronization, OneDrive URL verification, dynamic completion calculations, and exact enforcement of group leader submission permissions.

2. Technology Stack & Design System

Modern Frontend Engineering Stack

⚡ Core Framework & Tooling

- **React 18.3:** Functional components & custom reactive hooks
- **Vite 6.1:** Sub-second HMR and optimized production bundling
- **React Router DOM 6.28:** Nested routes, auth redirects, and guards

🎨 UI, Styling & Visuals

- **Tailwind CSS 3.4:** Custom Indigo/Navy SaaS theme tokens
- **Lucide React:** Semantic iconography across all workflows
- **Recharts 2.15:** Interactive course submission analytics

💾 State & Mock API Layer

- **Context API:** `AuthContext` and `DataContext`
- **Safe localStorage:** Auto-seeded DB with instant reset feature
- **Simulated Latency:** Realistic async promise responses

🔔 Feedback & Interactivity

- **React Hot Toast:** Rich notifications for all mutations
- **Confirmation Modals:** Guarding deletions, joins & submissions
- **Quick Role Switcher:** Instant 1-click toggle for evaluators

3. Demo Credentials & Pre-Seeded Data

Ready-to-Test Accounts with Simulated Real-World Data

ROLE	EMAIL ADDRESS	PASSWORD	SEEDED PERSONA & CAPABILITIES
Professor	**professor@joineazy.com**	`Professor@123`	Dr. Alistair Vance • 3 Active Courses (CS201, CS301, CS401) • Full Assignment CRUD &

ROLE	EMAIL ADDRESS	PASSWORD	SEEDED PERSONA & CAPABILITIES
			Tracking
Student	**student@joineazy.com**	`Student@123`	Alex Rivera (CS21B001) • Leader of *Alpha Devs* in CS401 • Member in CS301 • Unassigned in CS201
Student 2	**sarah.chen@joineazy.com**	`Student@123`	Sarah Chen (CS21B002) • Leader of *SQL Wizards* in CS301 • Member in CS401
Student 3	**priya.patel@joineazy.com**	`Student@123`	Priya Patel (CS21B004) • Leader of *ByteCrafters* in CS401

4. Core Workflows & Feature Implementation

Deep Dive into Verified Functional Requirements

👩‍🏫 4.1. Professor Assignment Management & Submission Tracking

Professors can create, edit, delete, search, filter, and sort assignments across all enrolled courses. Form validation guarantees non-empty titles, descriptions, deadlines, and legitimate OneDrive web URLs.

- ✓ **Dynamic Metrics:** Displays *"Submitted: X / Y, Pending: Z, Progress: %"* with animated progress bars.
- ✓ **Individual Submission Audit:** Lists every enrolled student with avatar, email, Submitted/Pending badge, and timestamp.
- ✓ **Group Submission Audit:** Shows team names, assigned leader, member lists, and exact submission timestamps.

👨‍🎓 4.2. Student Submission & Acknowledgement System

Assignments feature status indicators, time-remaining counters, and direct OneDrive repository access.

- ✓ **Individual Submission:** Clicking *"Yes, I have submitted"* opens a confirmation dialog. Confirmation records the exact timestamp, updates the UI immediately, displays *"Submitted on [Date & Time]"*, and prevents duplicate submissions.
- ✓ **Group Leader Privileges:** Only the group leader sees the *"Yes, I have submitted (as Group Leader)"* button. Submitting records the acknowledgement across all group members.
- ✓ **Group Member View:** Non-leader members see *"Acknowledged by [Leader Name] on [Date and Time]"* without duplicate submit buttons.

Exact Requirement Verification: When a student views a group assignment without being part of any group, the system displays the mandatory notice:

> **"You are not part of any group. Create or join one to submit this assignment."**

Along with the action button: **"Create or Join Group"** routing directly to the course group management screen.

👥 4.3. Group Management System

Enforces university group collaboration rules:

- ✓ **Single Group Rule:** A student can belong to only one group per course.
- ✓ **Automatic Leadership:** The creator automatically becomes the group leader.
- ✓ **Capacity Guard:** Groups have a maximum member capacity (e.g. 4/4 Full). Full groups cannot accept more members.
- ✓ **Safe Leave Flow:** Students can leave a group with safety confirmation; if a leader leaves, leadership is automatically passed to the next member.

5. Production Verification & Deployment

Build Integrity, Git History & Live Environment

🌐 Live Production URL

Vercel Link:
https://frontend-v3yg.vercel.app

- Configured with SPA rewrite rules in `vercel.json`.
- Zero 404 errors on deep route refreshes.
- Fast global CDN delivery.

📦 Git Repository

Repository URL:
https://github.com/214G1A05C2/joineasytask2.git

- Clean Git history with 57 files committed.
- Branch: `main`.
- Commit: `bb80efd`.

6. Quality Assurance & Evaluation Checklist

REQUIREMENT MODULE	IMPLEMENTATION DETAIL	TEST STATUS
Authentication	Role-based route protection, session persistence, inline error validation, 1-click demo fill	Passed (100%)
Professor Dashboard	Live dynamic KPI calculations, Recharts analytics, course cards, direct navigation	Passed (100%)
Assignment CRUD	Create, view, edit, delete with modals, OneDrive link validation, search & filters	Passed (100%)
Submission Tracking	Audit view for individual students & groups, exact timestamps, detail modals	Passed (100%)
Student Dashboard	Personalized greeting, 6 stat cards, status donut chart, enrolled course progress	Passed (100%)
Individual Submit	Confirmation dialog, timestamp capture, immediate UI update, duplicate prevention	Passed (100%)
Group Submit	Leader-only button, all-member sync, non-leader acknowledgement notice	Passed (100%)
Group Management	Create team (auto-leader), join open teams, capacity check, leave confirmation	Passed (100%)

REQUIREMENT MODULE	IMPLEMENTATION DETAIL	TEST STATUS
UI / Responsiveness	Tailwind CSS SaaS palette, collapsible mobile sidebar, toast alerts, skeletons	**Passed (100%)**